



Orientation + Pediatric Concepts

Pediatric Dentistry OMFS 721

Joseph Yun, DDS
Ian Marion, DMD

Mission: To advance the science and art of health through education, service, scholarship and social accountability

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General Info

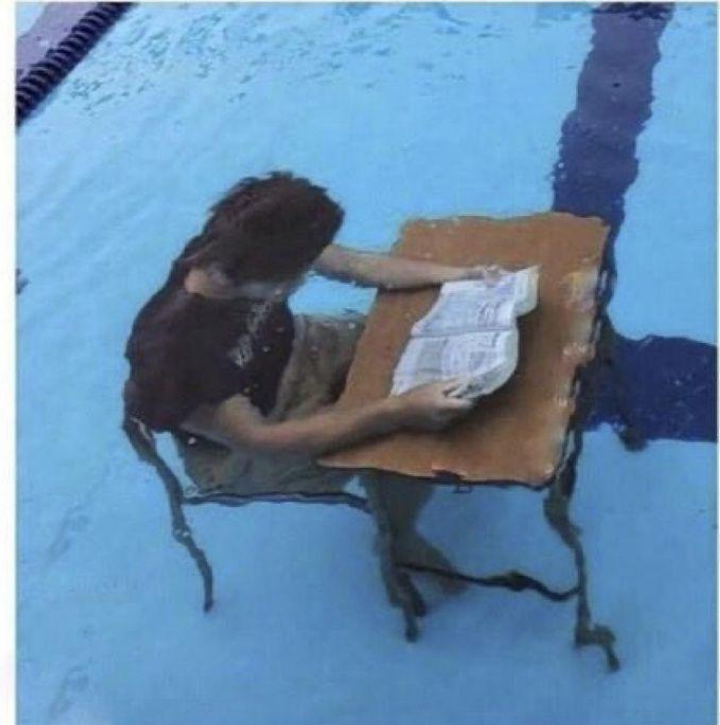
- Kwan-Ho.Yun@cnsu.edu
- Ian.marion@cnsu.edu



Class Expectations

- Class Design: We're all adults here!
 - Please be respectful
 - Scaffolded Learning
- Pre-class Readings available
- Small-group assignments
 - Welcome to finish them in class if lecture ends early!
- You get what you put in

Tip: study underwater so that no one can see you crying



Other General Info

- You're doctoral candidates!
- The class gets three warnings for breaches of professionalism
- Please show up on time.
- Do not disrupt your classmates



I can't believe I have to write this slide.

How to Prepare

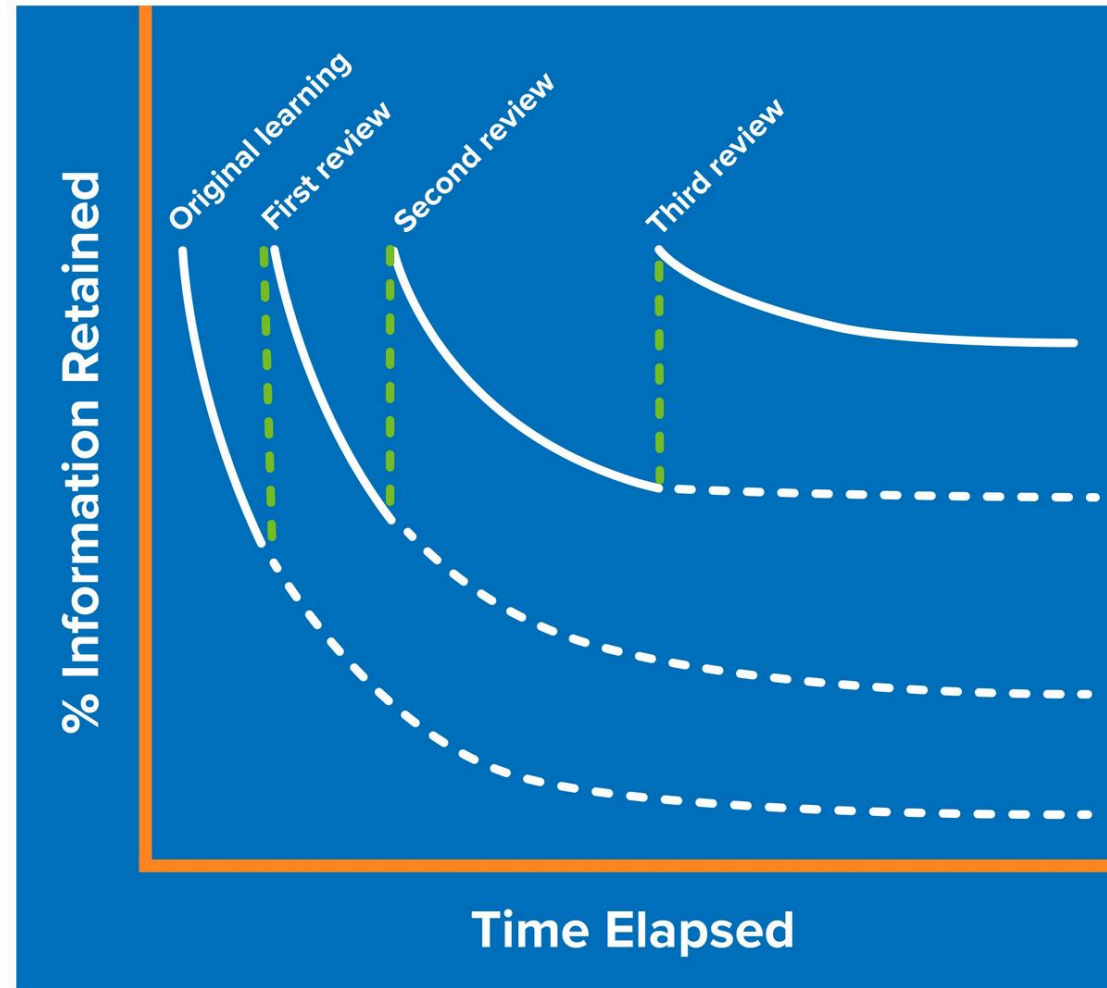
- Key concepts will be **highlighted** throughout the lecture.
 - Take notes!
- Get assistance if you need it!
 - Time available after classes
 - Ask about office hours
- Dr. Marion will provide summary sheets for his lectures



"Dr. Yun's office hours changed my life; before, I was lost and in the dark. Now, it's like the veil has lifted." – Nate

"He is a cool dude." - Jaspreet

How to Study: Ebbinghaus' Forgetting Curve



[Image Courtesy of Mindtools](#)

How is this class graded?

Assessment Method	Percentage Score
In-Class Assessment/Quiz	15%
Group Assignment/Small Group Activity	15%
Mid-Term Examination	25%
Final Examination	35%
Entrustable Professional Activities (EPA)	10%
Skill Assessments	
Total	100%

Assessments

- Quizzes – at the beginning of lectures – starting next week.
 - Material is on the **previous** class lectures
- Assignments (Problem Sets)
 - Based on completion
 - Completed in class

Course Outline

- 10 weeks of lectures (1 midterm)
- 3 weeks of lab experience
- 1 final exam

Assessments

- 1 Midterm
- 1 Cumulative Final with emphasis on new material but key concepts from first half included

What is Pediatric Dentistry?

- “Pediatric dentistry is an age-defined specialty that provides both primary and comprehensive preventive and therapeutic oral health care for infants and children through adolescence, including those with special health care needs.” – [American Academy of Pediatric Dentistry](#)
- Pediatric dentistry encompasses disciplines of:
 - Behavior guidance
 - Medically/developmentally/socially complex patients
 - Pharmacological management
 - Hospital dentistry
 - Caries prevention

So why do you care?



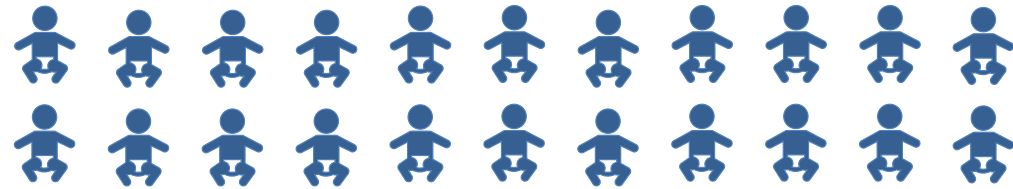
Pediatric Dentists in the US

9183

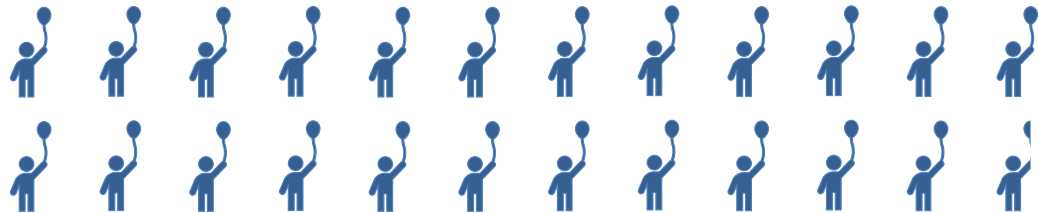


Children in the US

Ages 0–5: 22.0 million



Ages 6–11: 23.8 million



x100

Ages 12–17: 25.1 million



To see all children

- Pediatric dentists would need to see:
 - about 7622 patients annually
 - 60 patients every working day to see each patient twice a year

Therefore

Every child can not be
seen by a pediatric
dentist





Do you need to know Pediatrics as a General Practitioner

Yes

The vast majority of dental treatment for
children will be provided by:

You

“Family” Dentistry

- As a general practitioner you will see children
- You will see parents of children
- See who you can see
- Give advice if you cannot

Do baby teeth matter?

Yes

Baby Teeth



- Can cause pain, infection, and lasting damage if untreated
- Are essential for speech, mastication and growth

Do baby patients matter?

Yes

Young Patients

- Develop expectations about oral health and dentistry
- Establish habits that may dictate their oral health for life



With Great Dentistry Comes:

Great
Responsibility



Early Dental Experiences



Shape expectations
and behavior for
later dental
experiences

Dental Fear and Anxiety

- Half of **ADULTS** Experiencing Dental Fear had an Adverse Event in Childhood

As a practitioner seeing children



You will help shape
both the dental health
and attitudes of your
patients, whether you
intend to or not

If you hate children





There are reasons not to do, there is no
excuse not to know



What is Pediatric Dentistry?

Pediatric Dentistry Encompasses



Caries/Cavities





Habits



Malocclusion



Dental Anomalies



Special Needs



Sedation Dentistry



Hospital Care



General Anesthesia







Prevention and Conservative Treatment

What is Different?

- Pediatric dentistry is very *patient* driven
- Our procedures tend to be simple
- Our patients are often very complicated to handle
- We have to come up with creative ways to accomplish our goals



Most of All

Pediatric Dentistry is.....

FUN!





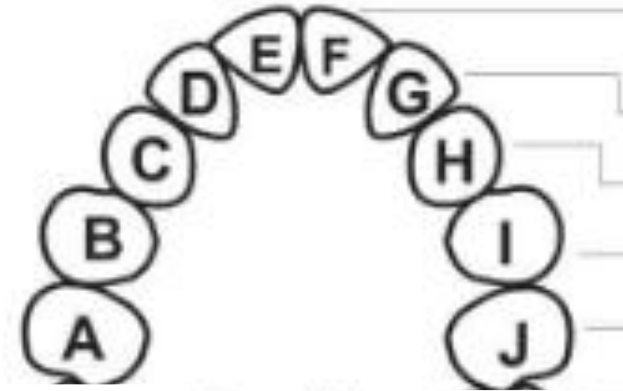


What is there to learn?

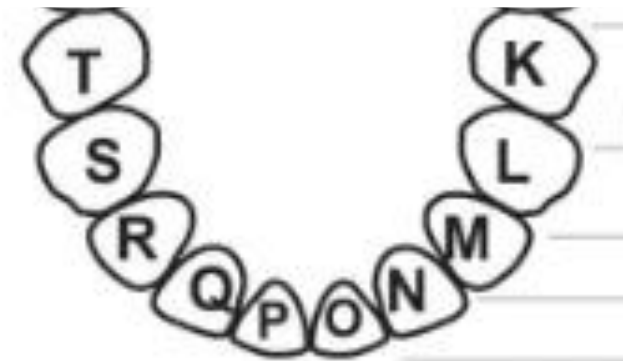
The Alphabet



Primary Teeth



Baby



Dental Home

- **The Dental Home**: Ongoing relationship between the dentist and patient, inclusive of all aspects of oral health care delivered in a comprehensive, accessible, coordinated way.
- Addresses anticipatory guidance and preventive, acute, and comprehensive oral health care.

What is the AAPD stance on dental homes?

- Should be no later than 12 months of age!
- Officially, when the first tooth erupts or by 12 months, whatever happens first.
- The goal is to see children early, before the problems occur

Early Childhood Caries (ECC)

- **Early Childhood Caries (ECC):** The AAPD defines early childhood caries as: the presence of 1 or more decayed (noncavitated or cavitated lesions), missing (due to caries), or filled tooth surfaces in any primary tooth in a child 71 months of age or

Severe Early Childhood Caries (S-ECC)

- In children younger than 3 years of age, any sign of smooth-surface
- From ages 3 through 5, 1 or more cavitated, missing (due to caries), or filled smooth surfaces in primary maxillary anterior teeth
- $A\ dmfs \geq (age + 1)$
 - ≥ 4 (age 3),
 - ≥ 5 (age 4),
 - ≥ 6 (age 5)

White Spot Lesions



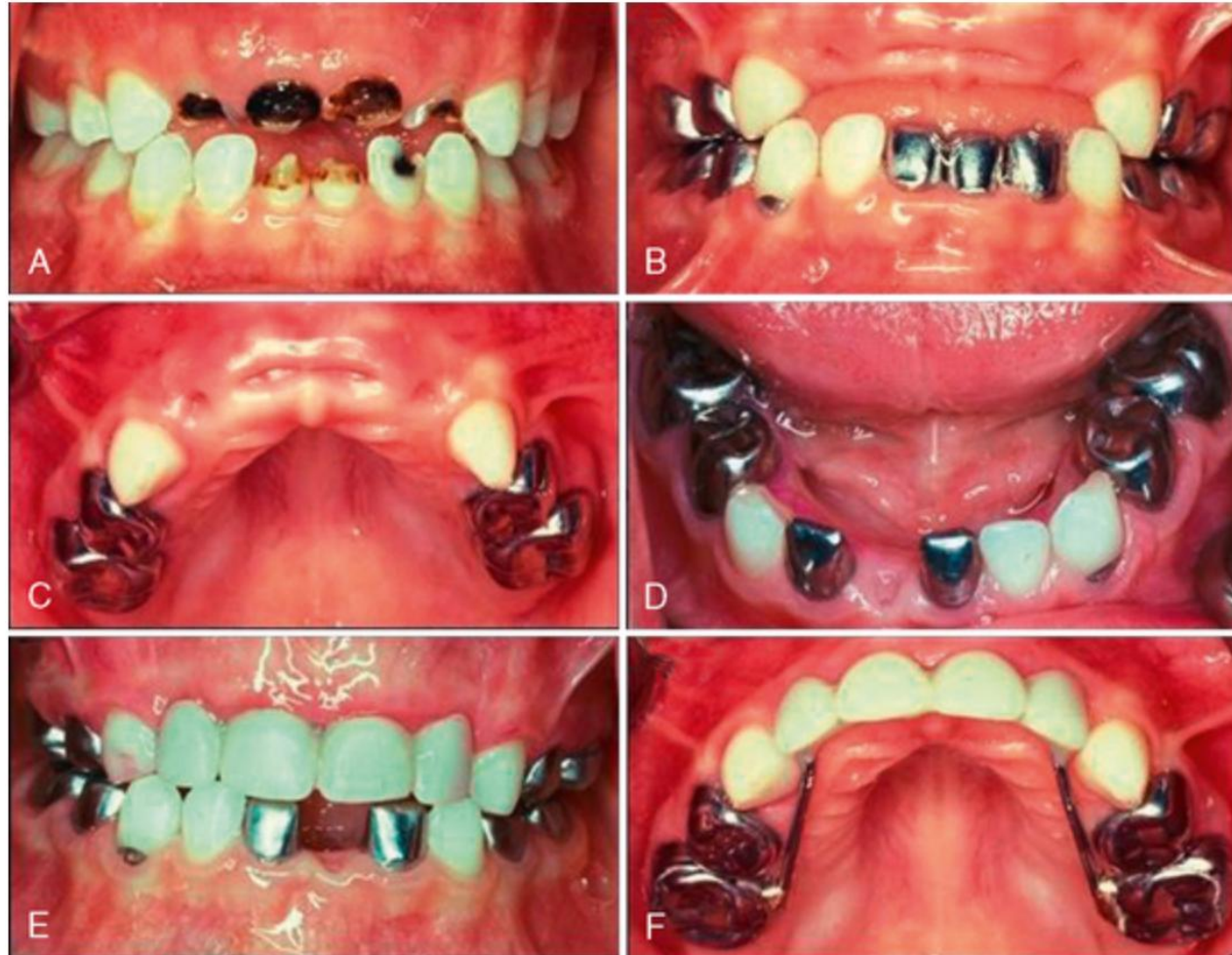
The white spots in primary teeth are decalcified areas—primary teeth are thinner and more transparent.

Why does ECC matter?

- Higher risk of new carious lesions (caries begets caries)
- Hospitalizations and emergency room visits
- High treatment costs
- Loss of school days (leading cause of missed school days);
diminished ability to learn
- Diminished oral health quality of life

dmfs vs DMFS

- dmfs refers to primary dentition
- DMFS refers to adult dentition



[Source: Pocket Dentistry](#)

Things you should review regularly

Hint hint

- Caries Risk Assessment
- Tooth Morphology (not discussed today)
 - Dental Eruption Dates/Patterns
 - Tooth Development Dates

Table 2. Caries-risk Assessment Form for 0-5 Year Olds^{59,60}
(For Dental Providers)

Factors	High Risk	Moderate Risk	Low Risk
Biological			
Mother/primary caregiver has active caries	Yes		
Parent/caregiver has low socioeconomic status	Yes		
Child has >3 between meal sugar-containing snacks or beverages per day	Yes		
Child is put to bed with a bottle containing natural or added sugar	Yes		
Child has special health care needs		Yes	
Child is a recent immigrant		Yes	
Protective			
Child receives optimally-fluoridated drinking water or fluoride supplements			Yes
Child has teeth brushed daily with fluoridated toothpaste			Yes
Child receives topical fluoride from health professional			Yes
Child has dental home/regular dental care			Yes
Clinical Findings			
Child has >1 decayed/missing/filled surfaces	Yes		
Child has active white spot lesions or enamel defects	Yes		
Child has elevated mutans streptococci levels	Yes		
Child has plaque on teeth		Yes	

Circling those conditions that apply to a specific patient helps the practitioner and parent understand the factors that contribute to or protect from caries. Risk assessment categorization of low, moderate, or high is based on preponderance of factors for the individual. However, clinical judgment may justify the use of one factor (e.g., frequent exposure to sugar-containing snacks or beverages, more than one dmfs) in determining overall risk.

Overall assessment of the child's dental caries risk: High ☐ Moderate ☐ Low ☐

Special Health Care Needs (SHCN)

- “Special health care needs include any physical, developmental, mental, sensory, behavioral, cognitive, or emotional impairment or limiting condition that requires medical management, health care intervention, and/or use of specialized services or programs. The condition may be congenital, developmental, or acquired through disease, trauma, or environmental cause and may impose limitations in performing daily self-maintenance activities or substantial limitations in a major life activity.”

Evaluating Med Hx

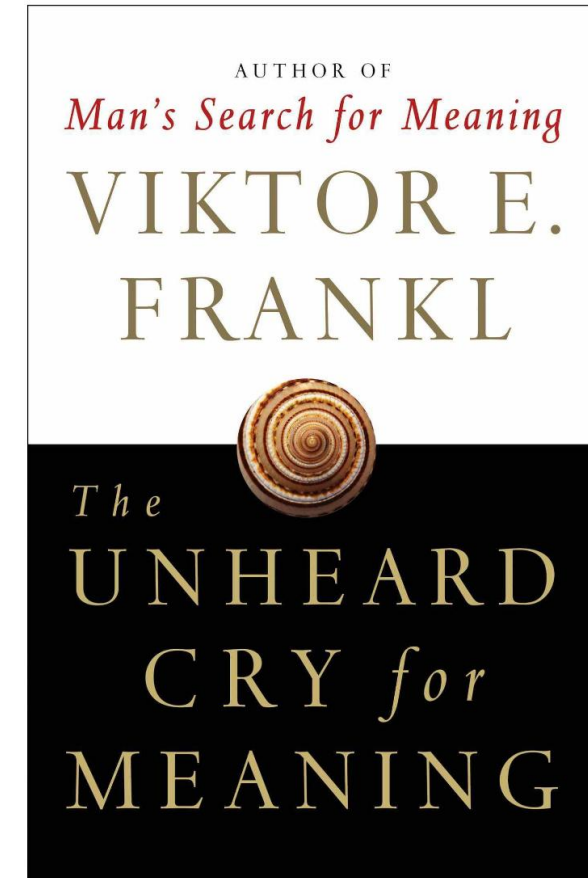
- American Society of Anesthesiologists (**ASA**) Classification

Classification	Description
I	Healthy patient
II	Mild to moderate systemic condition, WELL-CONTROLLED
III	Severe systemic disease but NOT life-threatening
IV	Severe disease w/ constant threat to life
V	Moribund patient not expected to survive within 24 hrs with or without operation
VI	Brain-dead patient on life-support for organ transplant

Behavior - Frankl Scale

- Developed in 1962

Appendix 2. FRANKL BEHAVIORAL RATING SCALE		
1	--	Definitely negative. Refusal of treatment, forceful crying, fearfulness, or any other overt evidence of extreme negativism.
2	-	Negative. Reluctance to accept treatment, uncooperative, some evidence of negative attitude but not pronounced (sullen, withdrawn).
3	+	Positive. Acceptance of treatment, cautious behavior at times, willingness to comply with the dentist, at times with reservation, but patient follows the dentist's directions cooperatively.
4	++	Definitely positive. Good rapport with the dentist, interest in the dental procedures, laughter and enjoyment.



"Unheard Cry" seems pretty apt for the dental school experience.

Apprehension

- **Apprehension:** panic response to the unknown
- Traditional, non-pharmacological behavior management very effective
- Most kids occupy this area



Fear

- **Fear:** panic response to a known event
 - Often occurs as a response to genuine trauma
- Typically requires advanced behavior management
- Older children more susceptible



Today's Activity

- In your tables each person share
 - Your comfort level dealing with kids on a 0-10 scale
 - Your experiences as a kid at the dentist
 - What part of dental care for kids you think will be most challenging
- No submission required